HDOC Day-69

Question: Write a recursive function to generate the first n Fibonacci numbers. Also count the number of recursive calls made.

Python Program

```
calls = 0

def fibonacci(n):
    global calls
    calls += 1
    if n <= 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)

n = int(input("Enter number of terms: "))

if n <= 0:
    print("Please enter a positive number.")
else:
    print("Fibonacci series:")
    for i in range(n):
        print(fibonacci(i), end=" ")
    print("\nTotal function calls:", calls)
    print("Recursive calls:", calls - n)
```

Sample Output

```
Enter number of terms: 5
Fibonacci series:
0 1 1 2 3
Total function calls: 19
Recursive calls: 14
```

Explanation

The Fibonacci series begins with 0 and 1. Each next term is the sum of the previous two terms. The fibonacci() function calls itself to calculate each term. The variable calls counts every invocation of the function. Since the program calls fibonacci() once for each of the n terms, subtracting n gives the number of recursive invocations made inside the function.

Result

The program successfully generates the first n Fibonacci numbers using recursion and counts the recursive calls.